

Auto Ticket Classification using Flow Designer

Project Documentation
School IT Helpdesk | ServiceNow

This project is built on ServiceNow, so each item of the MERN documentation template is mapped to its ServiceNow equivalent (React: platform UI, Node.js/Express: Flow Designer, MongoDB: custom table).

1. Introduction

- **Project Title:** Auto Ticket Classification using Flow Designer (School IT Helpdesk)
- **Team ID:** SWUID2026297489
- **Team Size:** 3

2. Project Overview

- **Purpose:** The school IT helpdesk receives daily requests from students and teachers (Wi-Fi, projector, password and slow-computer issues), and staff currently classify each one by hand. This project automatically classifies incidents at creation time using keywords in the short description. Goals: reduce manual effort, improve ticket routing, and keep the solution no-code and easy to maintain.

Features:

- Custom table **Incident Workflow** (u_incident_workflow) with auto-numbering (INC00500 onward, 5 digits)
- Dependent Category to Subcategory choice fields
- Keyword-based auto-classification with Flow Designer
- Automatic email acknowledgement to the caller
- Update set export (XML) for moving the solution to other instances

3. Architecture

Frontend (instead of React): The ServiceNow platform UI. The form was designed in Form Designer and shows Number, Caller, Category, Subcategory, Short description, Description, State, Assigned group and Assigned to.

Backend (instead of Node.js/Express): A Flow Designer flow named *Auto Classify School IT Tickets*.

- Trigger: Record Created on Incident Workflow, condition Category is empty
- If / Else If branches check the Short description with the Contains operator

- Update Record actions set Category and Subcategory
- Send Email action notifies the caller

Keyword in Short Description	Category	Subcategory
Wi-Fi / Network	Network	Wi-Fi
Projector	Hardware	Projector
Forgot password	Access	Forgot Password
Slow computer	Performance	Slow Computer

Database (instead of MongoDB): Table u_incident_workflow

Field	Type	Reference / Choices
Number	Auto Number	Prefix INC
Caller	Reference	sys_user
Category	Choice	Network, Hardware, Access, Performance
Subcategory	Choice (dependent on Category)	Wi-Fi, Projector, Forgot Password, Slow Computer
Short Description	String	-
Description	String	-
State	Choice	New, In progress, On hold, Resolved, Closed
Assigned Group	Reference	sys_user_group
Assigned to	Reference	sys_user

4. Setup Instructions

Prerequisites:

- A ServiceNow instance (for example a Personal Developer Instance) with admin access
- Flow Designer / Workflow Studio available
- Outbound email configured, or use the Emails log to preview messages
- A modern web browser

Installation - Option 1: Import the update set

System Update Sets, Retrieved Update Sets, Import Update Set from XML, upload the exported file, then Load, Preview and Commit.

Installation - Option 2: Build manually

- 1 Create the update set "Project Update Set" (Global, In progress) and make it current.
- 2 Create the table "Incident Workflow" (untick Create module, enable Auto-number with prefix INC).
- 3 Add the fields and choices through Form Designer.
- 4 On the Subcategory dictionary entry, enable Use dependent field with Category, then add the dependent value on each choice (Wi-Fi: network, Projector: hardware, Forgot Password: access, Slow Computer: performance).
- 5 Build the flow and click Activate.

5. Folder Structure

ServiceNow has no folders, so this maps to platform artifacts.

Client (UI side):

- Form layout and Form Designer view
- Field labels and choice lists
- Dictionary dependency settings

Server (logic side):

- Update Set: Project Update Set
- Table and fields: u_incident_workflow
- Flow: Auto Classify School IT Tickets
- Role: u_incident_workflow_user (created with the table's access controls)
- Email log: System Logs, Emails

6. Running the Application

- **Frontend:** Open All menus, Incident Workflow, New. Enter the Caller and Short description, then submit.
- **Backend:** Open Flow Designer, open the flow and click Activate. It then runs automatically on every new record. Check executions under Operations, Flow Execution.

7. API Documentation

No custom API was written because the logic runs inside Flow Designer. ServiceNow's standard Table API can still create tickets, and the flow triggers on them too.

Create a ticket:

```
POST /api/now/table/u_incident_workflow

Body (example): { "u_caller": "<sys_id>", "u_short_description": "Projector not turning on" }
```

Response: 201 Created with the record, including the generated number (for example INC00512). Category = Hardware and Subcategory = Projector are set once the flow completes.

Read tickets:

```
GET /api/now/table/u_incident_workflow
```

Note: verify the exact field names (u_...) on your table; the standard naming convention is assumed here.

8. Authentication

- Authentication is handled by the ServiceNow platform login. Sessions are platform-managed, with no custom tokens.
- Authorization uses the role u_incident_workflow_user, created with the table's access controls (ACLs), so only authorised users can read or modify ticket data. The admin role has full access.

9. User Interface

Screenshots to include from the project:

- New record form with Category and Subcategory empty
- Dependent subcategory dropdowns (Network: Wi-Fi, Hardware: Projector, Access: Forgot Password, Performance: Slow Computer)
- Flow Designer trigger and the If / Else If branches

- Update Record and Send Email action screens

10. Testing

- **Strategy:** Functional end-to-end testing by creating records manually and checking the results, plus data validation checks.
- **Data validation:** mandatory fields captured, no duplicate auto-numbers, Category and Subcategory stored correctly, reference fields resolving to user and group tables.
- **Email check:** System Logs, Emails, search for the subject "Your Request for the issue has been submitted", then Preview Email.

#	Short Description	Expected Result	Outcome
1	Issue with Network (INC00511)	Network / Wi-Fi, email sent	Passed
2	Projector not turning on (INC00512)	Hardware / Projector, email sent	Passed

11. Screenshots or Demo

Include screenshots of INC00511 and INC00512 after classification, the Emails log with recipients and subject, the email preview, and the update set download (sys_remote_update_set_...xml, 182 KB).

12. Known Issues

- Only exact keyword matches are classified. The flow checks "Forgot password" and "Slow computer", not "Login", "Hanging" or "Slow" alone.
- "WiFi" without the hyphen will not match the "Wi-Fi" condition.
- Tickets that match no keyword stay uncategorised, because there is no default Else branch.
- Mixed descriptions such as "Projector Wi-Fi not working" may be misclassified, since the first matching branch wins.
- The email text needs a grammar fix and should include the ticket number.

13. Future Enhancements

- Add the missing keywords (Login, Hanging, WiFi) and a default Else branch that routes to manual triage
- Auto-assign the Assigned group by category (Network team, Hardware team, and so on)
- Set priority from words like "urgent" or "exam"
- Send a second email when the State changes to Resolved
- Store keywords in a lookup or decision table so staff can edit them without touching the flow
- Add SLA tracking and Predictive Intelligence for smarter classification